

Robotics Assignment: CAD-Based Robot Structure Identification and Motion Demonstration

Course Level

3rd-Year Computer Engineering Students

Assignment Type

Group Assignment

CA Weight

This assignment carries **16 marks out of 40 Continuous Assessment (CA) marks**.

Main Software

- SolidWorks
 - MATLAB
 - Simulink
 - Simscape Multibody, where available
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1. Assignment Overview

In this assignment, each group will select one robot CAD model provided as a ZIP file. Each ZIP file contains a SolidWorks robot assembly and its related part files.

Each group will inspect the selected robot assembly, identify the main robot parts, determine the robot type, estimate the degrees of freedom, and prepare a simple robot motion demonstration. The assignment focuses on helping students understand how a robot CAD assembly can be interpreted as a practical robotics system made of a base, links, joints, moving parts, and an end-effector.

The assignment also introduces students to the process of preparing a CAD robot model for motion demonstration using MATLAB, Simulink, or Simscape Multibody.

2. Group Selection Rule

Each group must choose only one ZIP file.

- A ZIP file selected by one group must not be selected by another group.

This means that every group must work on a different robot model. Groups that select the same ZIP file must resolve the duplication before starting the assignment. If duplication

remains, the affected groups may be required to change their robot model or may lose marks for non-compliance.

3. Available Robot ZIP Files

The following robot ZIP files are available for selection:

No.	ZIP File Name	Likely Robot Type
1	5-axis-robotic-arm-2.snapshot.1.zip	5-axis robotic arm
2	6-axis-robot-6.snapshot.3.zip	6-axis robot
3	articulated-robot-4.snapshot.1.zip	Articulated robot
4	cartesian-manipulator-1.snapshot.3.zip	Cartesian manipulator
5	cylindrical-manipulator-1.snapshot.1.zip	Cylindrical manipulator
6	denso-4-axis-scara-robots-hs45452-1.snapshot.2.zip	4-axis SCARA robot
7	epson-prosix-vt6-a901s-6-axis-robot-1.snapshot.1.zip	6-axis industrial robot
8	industrial-robo-hand-1.snapshot.2.zip	Robotic hand or gripper
9	manipulator-cylindrical-1.snapshot.3.zip	Cylindrical manipulator
10	mini-robot-scara-mg995-with-pneumatic-suction-cup-1.snapshot.7.zip	Mini SCARA robot with suction cup
11	robot_arm_4-dof-1.snapshot.2.zip	4-DOF robotic arm
12	robotic-arm-409.snapshot.3.zip	Robotic arm
13	robotic-arm-abb-design-1.snapshot.3.zip	ABB-style robotic arm
14	robotic-arm-industrial-cad-model-1.snapshot.2.zip	Industrial robotic arm
15	robotic-arm-t310-1.snapshot.2.zip	Robotic arm
16	robotic-manipulator-arm-assembly-1.snapshot.1.zip	Robotic manipulator arm
17	scara-epson-t3-1.snapshot.2.zip	Epson SCARA robot
18	scara-robot-arm-2.snapshot.8.zip	SCARA robot
19	scara-robot-arm-8.snapshot.1.zip	SCARA robot

4. Assignment Objectives

By the end of this assignment, each group should be able to:

1. Open and inspect a SolidWorks robot assembly.
 2. Identify the base, links, joints, end-effector, and moving parts of a robot.
 3. Determine the robot type and estimate its degrees of freedom.
 4. Prepare or import the robot model into MATLAB/Simulink where possible.
 5. Design a simple robot motion trajectory.
 6. Demonstrate basic robot movement using screenshots, plots, or simulation.
 7. Present the robot structure, motion results, and limitations clearly.
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5. Group Tasks

Each group must complete all the tasks below using its selected ZIP file.

Task 1: CAD Assembly Inspection

Each group should extract the selected ZIP file and open the SolidWorks assembly.

The group should identify:

- main assembly file, such as .SLDASM
- part files, such as .SLDPRT
- base
- links
- moving parts
- fixed parts
- end-effector
- motors or actuator positions, where visible
- possible joint locations

Expected Output

The report should include:

- screenshot of the complete robot assembly
- list of the main files found in the ZIP folder
- labelled screenshot showing the main robot parts
- table showing important parts and their functions
- brief explanation of how the robot is mechanically assembled

Suggested Table

Part/File Name	Physical Part	Function in the Robot
Example: A.SLDASM	Main assembly	Contains the complete robot model
Example: B.SLDPRT	Base/link/motor mount	Supports or connects robot parts
Example: C1.SLDPRT	Link or joint part	Transfers motion to another part

Task 2: Robot Type and Degrees of Freedom Identification

Each group should identify the type of robot selected.

Possible robot types include:

- Cartesian robot
- cylindrical robot
- SCARA robot
- articulated robot
- 4-DOF robotic arm
- 5-axis robot
- 6-axis robot
- robotic hand or gripper

Each group should also estimate the number of degrees of freedom.

A degree of freedom is an independent motion of the robot. It may be a rotation, a linear movement, or an opening/closing movement in the case of a gripper.

Expected Output

The report should include:

Item	Description
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Selected ZIP file	Name of the selected robot ZIP file
Robot type	Cartesian, cylindrical, SCARA, articulated, etc.
Estimated DOF	Number of independent robot motions
Main application	Pick-and-place, handling, assembly, gripping, etc.
Reason for classification	Explanation based on the observed structure

Task 3: Link and Joint Identification

Each group should identify the links and joints of the selected robot.

A link is a rigid part of the robot. A joint is a connection that allows motion between two robot parts.

For each joint, describe:

- connected parts
- joint type
- motion direction
- possible axis of rotation or translation

Common joint types include:

Joint Type	Meaning
Revolute joint	Allows rotation about one axis
Prismatic joint	Allows linear motion along one axis
Fixed joint	Does not allow relative motion
Gripper joint	Allows opening and closing of the end-effector

Expected Output

The report should include a joint identification table.

Joint Number	Connected Parts	Joint Type	Motion Description	Axis/Direction
Joint 1	Base to Link 1	Revolute	Rotates the first arm link	About vertical axis

Joint 2	Link 1 to Link 2	Revolute	Rotates the second link	About joint axis
Joint 3	Link 2 to Link 3	Prismatic/Revolute	Moves or rotates the next link	Along or around selected axis
Joint 4	Link 3 to End-effector	Fixed/Revolute/Gripper	Positions or operates the tool	Tool direction

The number of rows should depend on the selected robot.

Task 4: Robot Motion Reference Points

Each group should define the important positions that the robot will use during the motion demonstration.

The group does not need to develop a complete coordinate-frame system. Instead, the group should define practical motion reference points using screenshots, labels, or simple position descriptions.

Examples of motion reference points include:

- home position
- object position
- pick position
- place position
- end position
- gripper open position
- gripper closed position

Expected Output

The report should include:

- screenshot of the robot at home position
- screenshot or sketch showing the target position
- description of the motion path
- table of motion reference points

Suggested Table

Reference Point	Description	Robot Condition
Home position	Starting position of the robot	Arm at rest
Object position	Position where the robot reaches the object	End-effector above object
Pick position	Position where the robot picks the object	End-effector lowered or gripper closed
Place position	Position where the robot moves the object	End-effector above destination
Final position	End of the movement	Robot stops or returns home

Task 5: Basic Robot Motion Description

Each group should describe how the selected robot moves from one position to another.

The group does not need to derive full forward kinematics equations. Instead, the group should explain the motion using joint movements and robot actions.

The explanation should answer the following questions:

1. Which joint moves first?
2. Which part of the robot rotates or translates?
3. How does the end-effector move?
4. What happens at the pick or target position?
5. What happens at the final position?

Expected Output

The report should include:

- simple motion description
- joint movement table
- explanation of how the end-effector reaches the target
- screenshots or sketches showing the motion sequence

Suggested Table

Step	Robot Action	Joint/Part Involved	Expected Effect
1	Start from home position	All joints	Robot begins from rest
2	Rotate or extend toward target	Joint 1/Link 1	Robot turns or moves toward object
3	Position end-effector	Joint 2/Joint 3	Tool reaches the working area
4	Pick or touch target	End-effector/gripper	Object is picked or target is reached
5	Move to final position	Main joints	Robot completes the task

Task 6: MATLAB or Simulink Multibody Preparation

Each group should attempt to prepare the robot model for MATLAB/Simulink simulation.

Possible activities include:

- exporting the SolidWorks assembly for Simscape Multibody
- importing the robot model into Simulink
- checking whether bodies and joints are correctly detected
- correcting missing joints
- defining suitable joint blocks
- connecting the robot base properly to the world frame
- testing simple joint motion

Expected Output

The report should include:

- screenshot of the imported model
- list of detected bodies
- list of detected joints
- explanation of any corrections made
- screenshot of a successful motion test

Important Note

If the import does not work correctly, the group should not stop. Instead, the group should explain:

- what worked
- what failed
- possible reason for the problem
- how the robot could be manually corrected or rebuilt in Simulink Multibody

This explanation will still be considered during assessment.

Task 7: Robot Motion Demonstration

Each group must design a simple motion task for the selected robot.

The motion may be:

- pick-and-place motion
- move from home position to target position
- end-effector positioning task
- gripper opening and closing
- arm extension and retraction
- joint rotation through selected angles

The motion should be completed within:

$$T_{(max)} = 2 \text{ s}$$

Example Motion Sequence

Time	Robot Action
0.0 s	Robot starts from home position
0.5 s	Robot moves toward the object or target
1.0 s	Robot reaches the pick or target position
1.5 s	Robot moves toward the place or final position

2.0 s	Robot releases, stops, or returns to a safe position
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Expected Output

The report should include:

- motion objective
- joint-space trajectory table
- plots of joint motion versus time, where possible
- screenshots of robot positions
- explanation of what the robot does during the 2-second motion

Suggested Joint Trajectory Table

Time	Joint 1	Joint 2	Joint 3	Joint 4	End-Effector State
0.0 s	0°	0°	0 mm/0°	0°	Open/Home
0.5 s	20°	15°	Up	0°	Moving to object
1.0 s	35°	25°	Down	0°	Pick or target position
1.5 s	-20°	30°	Up	0°	Moving to final position
2.0 s	-30°	10°	Down/Up	0°	Release/Stop

The exact joint values must be selected based on the robot model chosen by the group.

Task 8: Workspace or Reachability Observation

Each group should observe the working range of the selected robot.

The workspace is the region that the robot end-effector can reach.

Students may do this by:

- moving the robot joints manually in SolidWorks
- changing joint values in MATLAB/Simulink
- recording several end-effector positions
- sketching the reachable area
- identifying reachable and unreachable target points

Expected Output

The report should include:

- workspace sketch or screenshot
 - explanation of maximum reach
 - explanation of motion limitations
 - discussion of which tasks the robot can perform
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Task 9: Practical Applications and Limitations

Each group should explain where the selected robot can be used.

Examples:

Robot Type	Possible Application
Cartesian robot	3D printing, CNC positioning, material handling
Cylindrical robot	Machine loading, packaging, vertical handling
SCARA robot	Fast pick-and-place, assembly, sorting
Articulated robot	Welding, painting, handling, inspection
6-axis robot	Industrial automation and flexible manipulation
Robotic hand	Gripping, object handling, prosthetic or service robot studies

Each group should also discuss limitations such as:

- limited reach
 - limited payload
 - difficult joint control
 - collision risk
 - missing actuator details in the CAD model
 - incomplete or unclear CAD assembly constraints
 - simulation import problems
 - limited available information from the downloaded CAD files
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6. Report Structure

Each group must submit a report using the following structure:

1. Cover Page
2. Introduction
3. Selected Robot ZIP File
4. CAD Assembly Inspection
5. Robot Type and DOF Identification
6. Link and Joint Identification
7. Robot Motion Reference Points
8. Basic Robot Motion Description
9. MATLAB/Simulink Multibody Preparation
10. Robot Motion Demonstration
11. Workspace or Reachability Observation
12. Practical Applications and Limitations
13. Conclusion
14. References
15. Appendices

7. PowerPoint Presentation Structure

Each group must prepare a short presentation using the following structure:

1. Title slide
2. Selected robot ZIP file
3. Robot assembly overview
4. Robot type and estimated DOF
5. Main links and joints
6. Motion reference points
7. Basic robot motion description
8. MATLAB/Simulink preparation or simulation result

9. Robot motion demonstration
10. Workspace or reachability observation
11. Applications and limitations
12. Conclusion

8. Submission Requirements

Each group must submit:

1. Report in PDF format
2. Report in DOCX format
3. PowerPoint presentation
4. Screenshots of the SolidWorks assembly
5. MATLAB/Simulink files, where applicable
6. Generated plots or trajectory files, where applicable
7. Short video or screenshots of robot motion, where possible

9. Marking Rubric

This assignment carries **16 marks out of 40 CA marks**.

Assessment Item	Marks
Correct selection and use of a unique ZIP file	1.0
CAD assembly inspection and part identification	2.5
Robot type and DOF identification	1.5
Link and joint identification	2.5
Robot motion reference points	1.5
Basic robot motion description	1.5
MATLAB/Simulink preparation or simulation attempt	2.0
Robot motion demonstration and trajectory	1.5
Workspace or reachability observation	1.0

Report quality and presentation	1.0
Total	16.0

10. Important Instructions

1. Each group must choose a different ZIP file.
 2. No two groups are allowed to use the same ZIP file.
 3. The selected robot must be clearly stated in the report and presentation.
 4. The group must include screenshots from SolidWorks.
 5. The group must explain both successful and unsuccessful simulation steps.
 6. The assignment should show understanding, not only screenshots.
 7. All group members must understand the selected robot and be ready to answer questions.
 8. Groups should not wait until the simulation works perfectly before writing the report. The process, observations, problems, and corrections are part of the assignment.
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11. Expected Learning Outcome

After completing this assignment, students should be able to explain how a robot CAD assembly can be interpreted as a robotics system made of a base, links, joints, moving parts, and an end-effector.

Students should also be able to describe robot movement, estimate robot degrees of freedom, prepare simple robot motion demonstrations, and explain the practical challenges involved in using CAD models for robotics simulation.